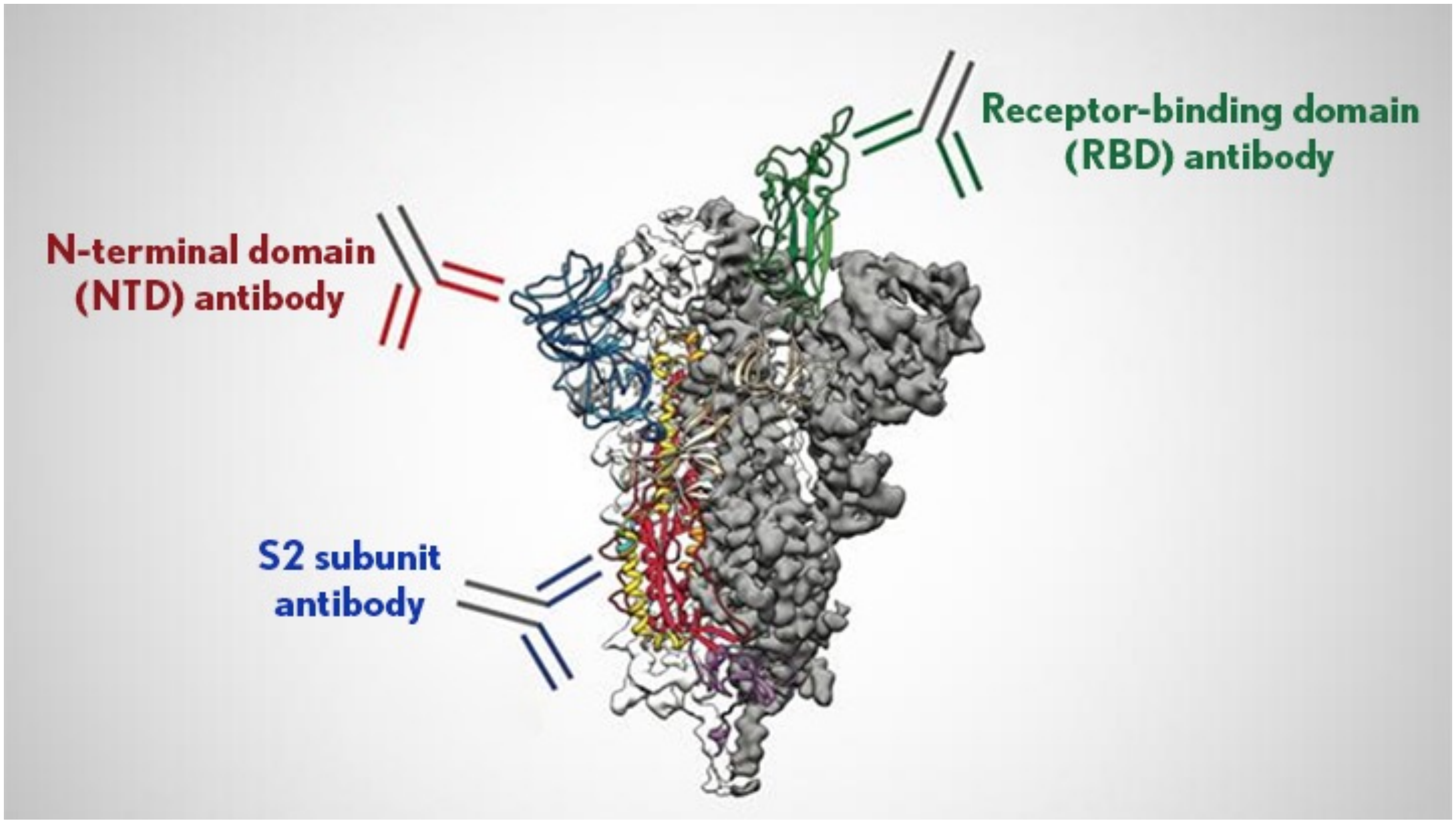
Immunogenetics

Readings: Pierce Ch 22.6 (Skip Major Histo… & Gene and Organ trans….)

Textbook problems: 8, 30, 34



People who recovered from mild COVID-19 *infections produced antibodies circulating in their blood that target three different parts of the coronavirus's spike protein (gray). Credit: University of Texas at Austin (https://directorsblog.nih.gov/2021/05/18/human-antibodies-target-many-parts-of-coronavirus-spike-protein/)*

Few definitions

Antigen: any molecule (usually a protein) that elicits an immune response

Antibody: (also called immunoglobulins) proteins (present in the blood and other body fluids) that bind to antigens, marking them for destruction by phagocytic cells.

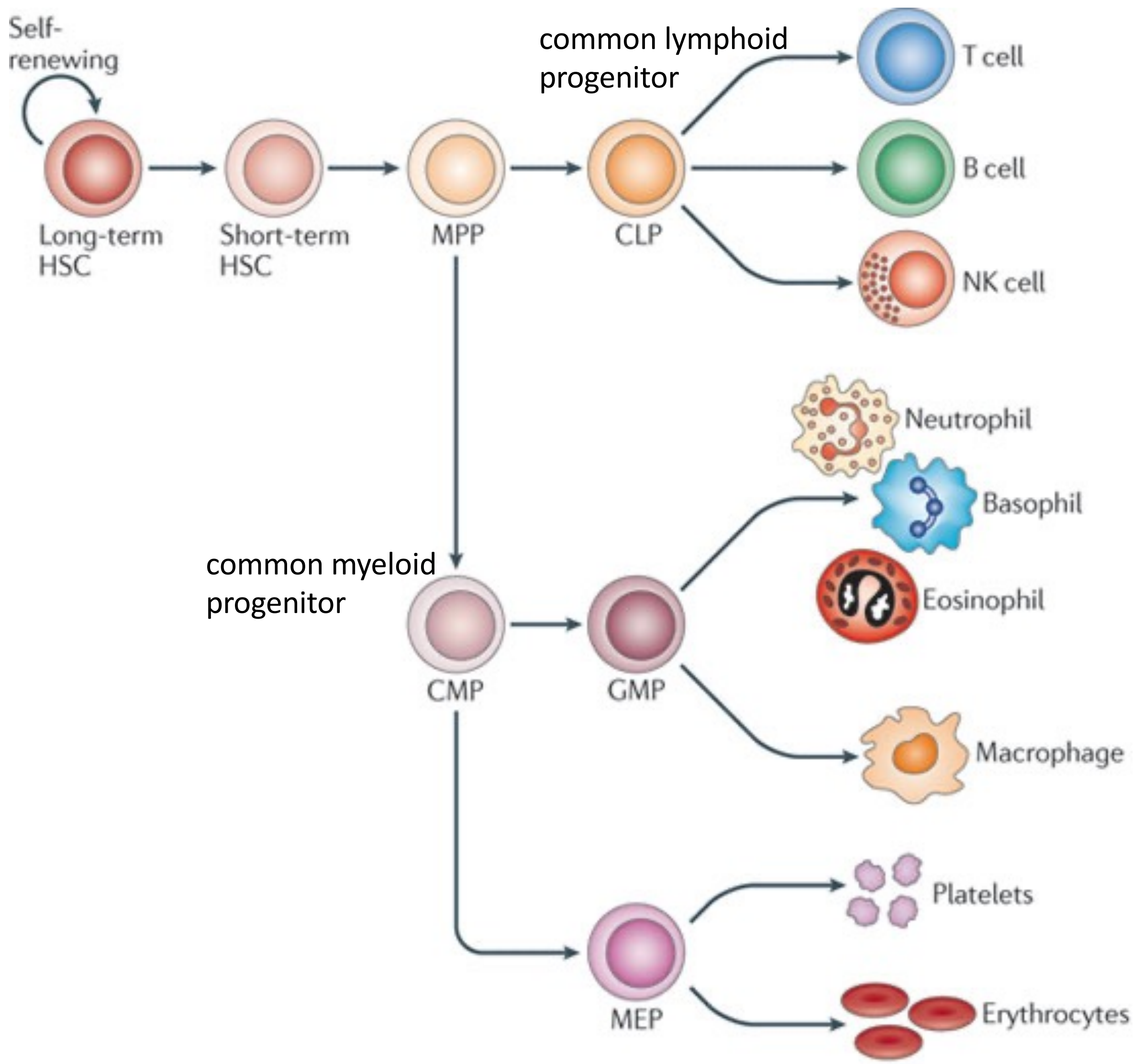
Autoimmune diseases: an immune reaction against its own antigens (proteins).

Adaptive Immune response
(antigen-specific immune response)

Humoral immunity: production and secretion of antibodies by a specialized lymphocytes (white cells) called **B cells**.

Cellular immunity: Specialized lymphocytes called **T cells** produce T-cell receptors that recognize and bind antigens found only on the surface of the body's own cells.

The hematopoietic tree



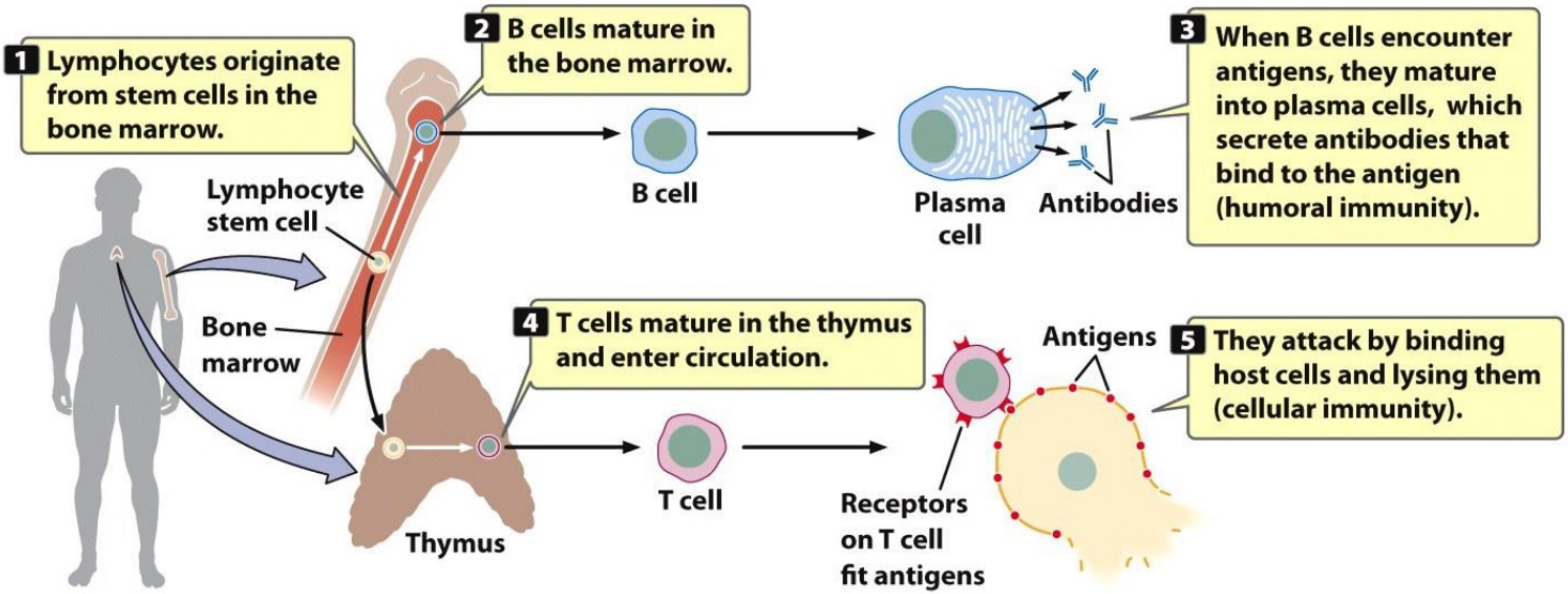
Nature Reviews | Immunology

Adaptive immunity
(antigen-specific defense)

Innate immunity
(non-specific defense but quick response)

Development of humoral (B-cell) and Cellular (T-cell) immunity

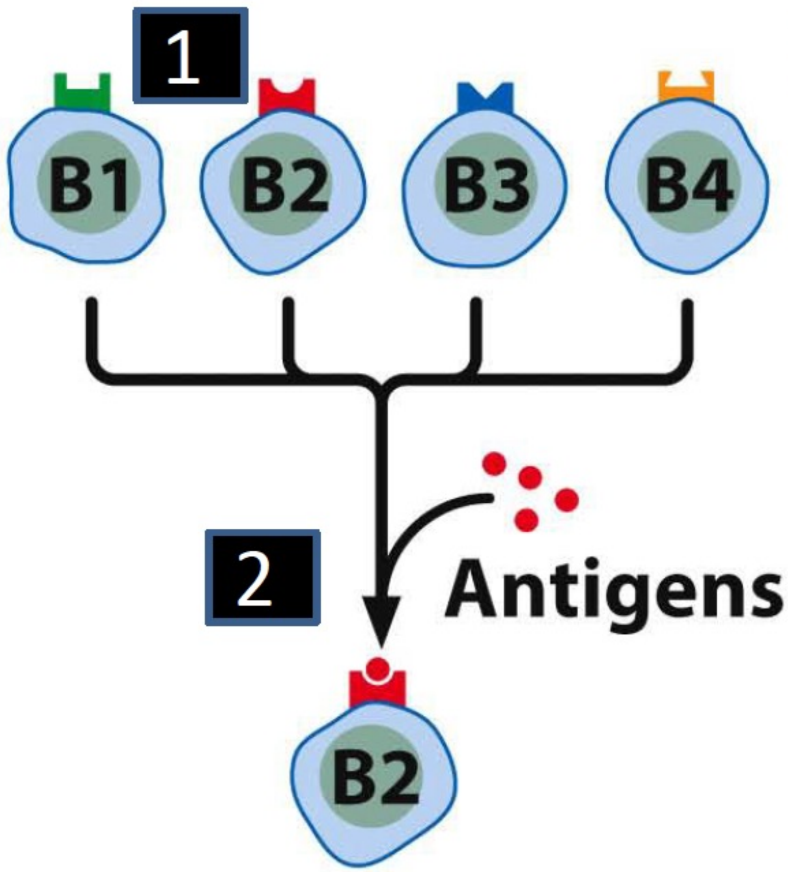
Autoreactive B cell and T cell are ELIMINATED during development



Pierce. 22.20

Each B cell and T cell recognize a unique <u>non-self antigen</u> that can elicit immune response (<u>diversity</u> in antibody and T cell receptor)

Clonal selection of B-cell that binds to a specific antigen



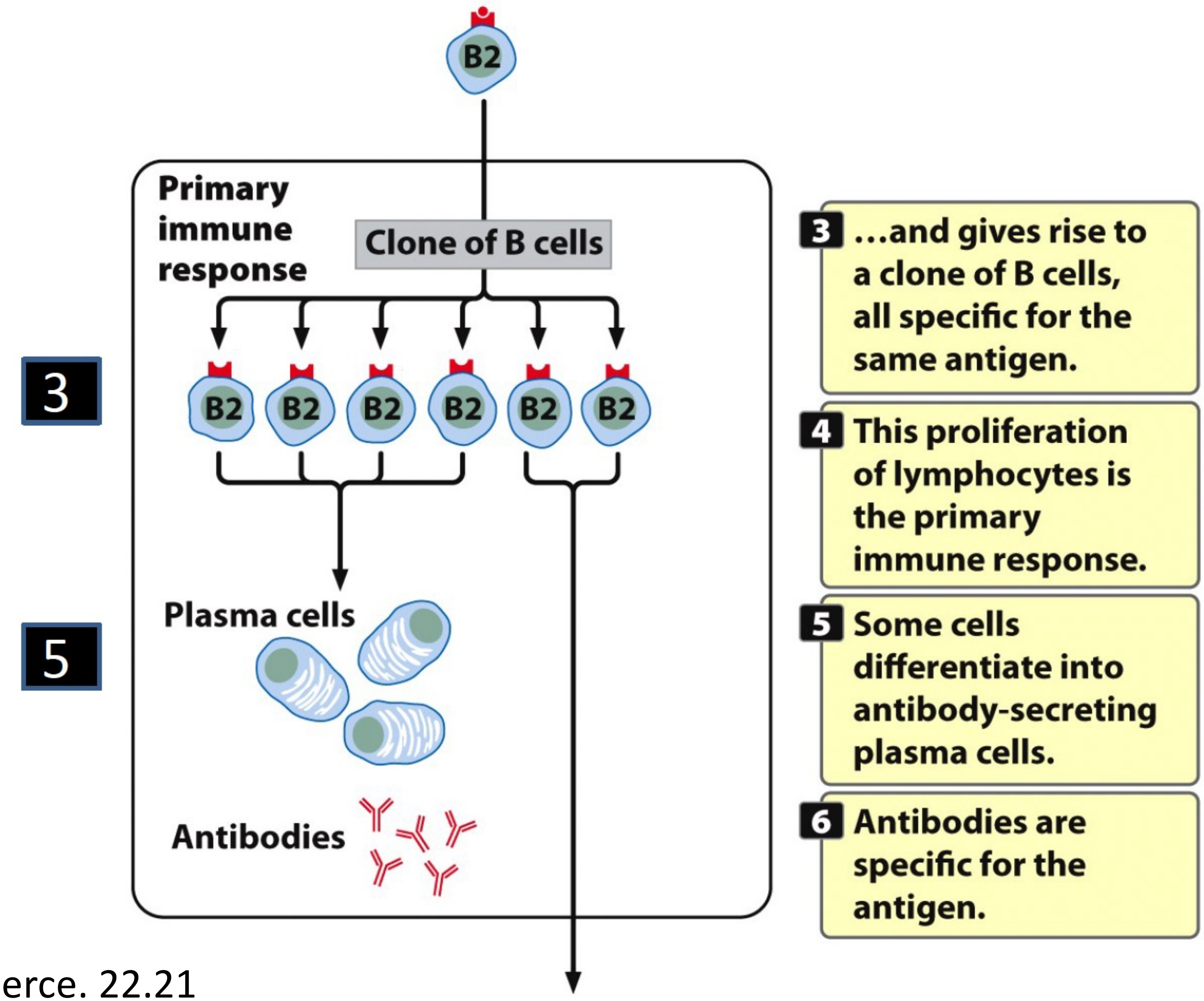
1 In a large pool of B lymphocytes, each is specific for one antigen.

(Non-self)

2 When an antigen binds to a B cell, the B cell divides…

Pierce. 22.21

Clonal expansion of the selected B-cell



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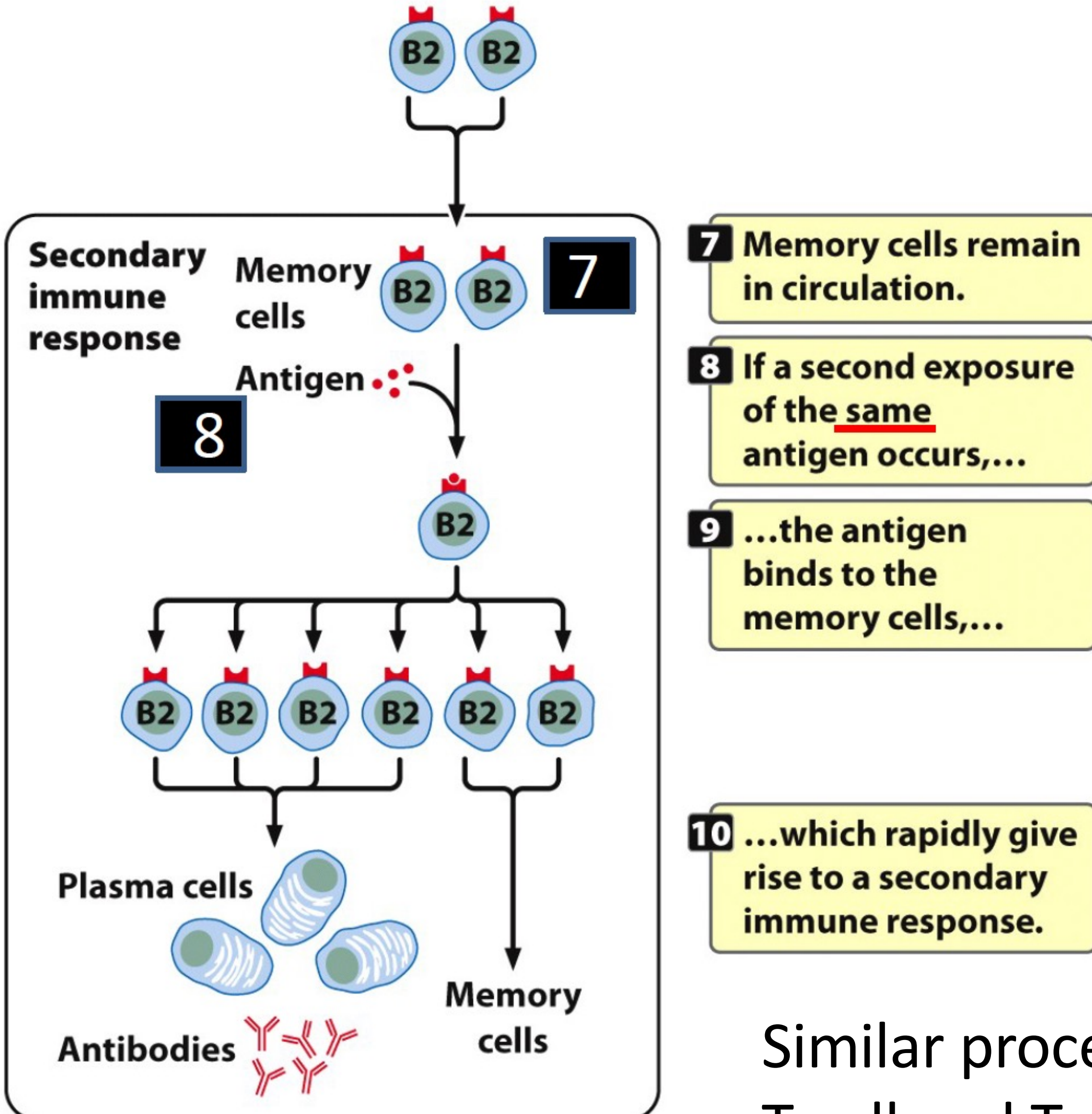
Secondary response by memory B cells that have been activated



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Similar process occurs for T cell and T cell receptor

The second immune response is the reason why vaccination works

The **active agent** (the antigen) of a vaccine:

Intact but inactivated (non-infective) pathogen

Attenuated (reduced infectivity) forms of the pathogen

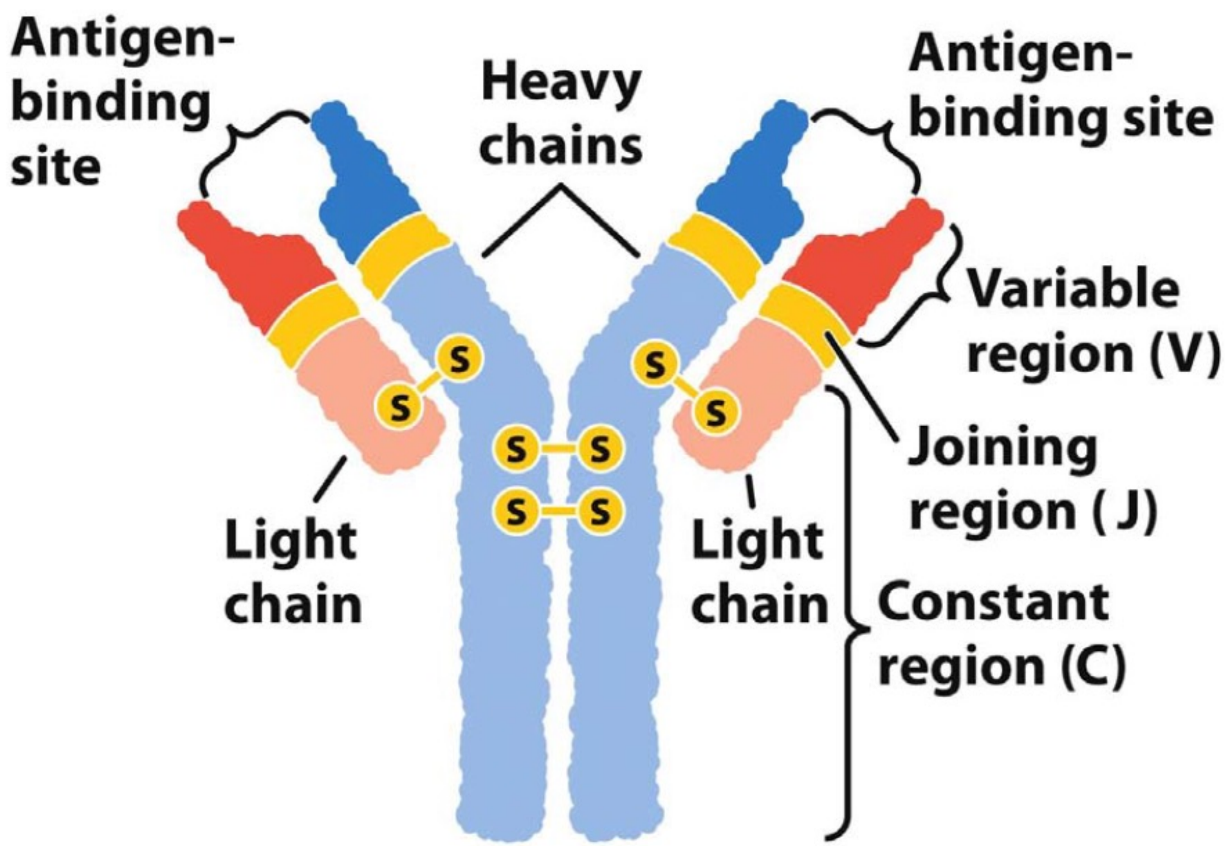
Purified components of the pathogen
that have been found to be highly immunogenic

The vaccine thus induces the primary immune response,
generating the memory cells that are ready
when the (second) infection happens.

Conventional methods (eg. culturing virus) take long time to develop. The next generation vaccines, such as RNA vaccine, use host cells to produce antigen, which is much faster.

Immunoglobulins: the proteins that protect us

two identical light chains and two identical heavy chains



Pierce. 22.22

Light chains and heavy chains

Light chains: two types (kappa and lambda) encoded on different chromosomes

Segments V, J, and C

(kappa/kappa or lambda/lambda)

Heavy chains: five types

(alpha, delta, gamma, epsilon or mu)

Segments V, D, J, and C

Each type of light chain can potentially **combine** with each type of heavy chain

B cells produce a unique light and heavy chain combination that can recognize a specific antigen

It is estimated that the human genome has 20,000 to 25,000 protein coding genes

However, humans have the capacity to produce 10^{11} antibodies with different specificities.

Three ways of generating the diversity:

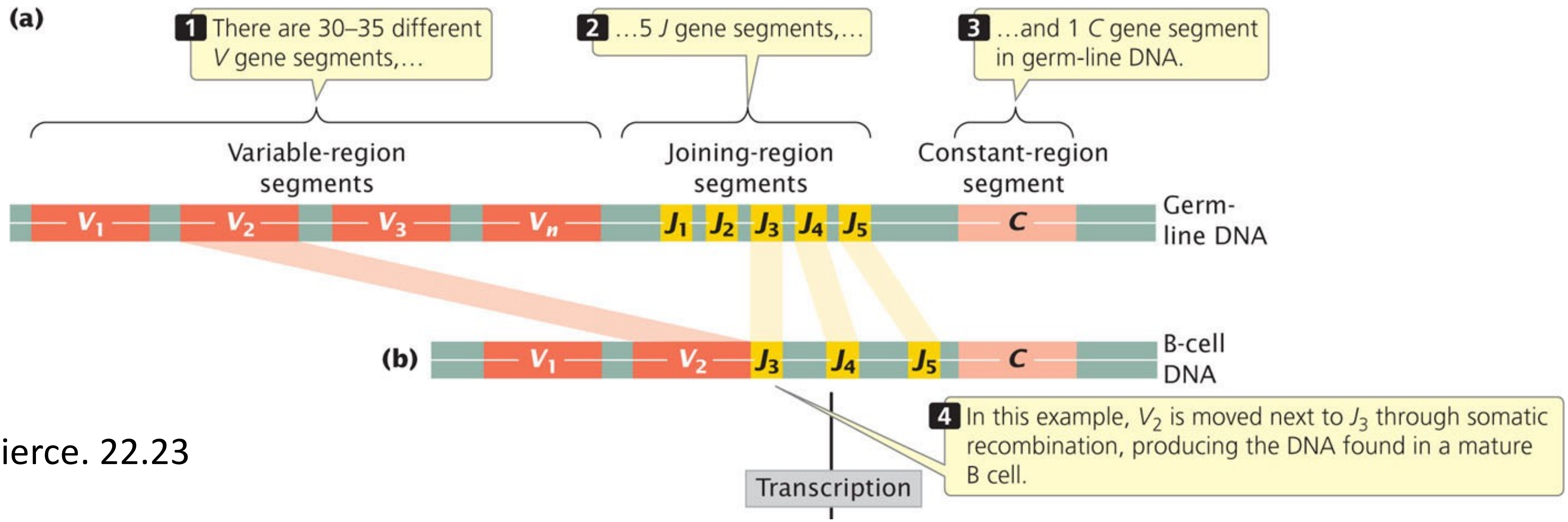
Somatic recombination

Junctional diversity

Somatic hypermutation

Somatic recombination between V and J gene segments of kappa light chain

During lymphocyte development……



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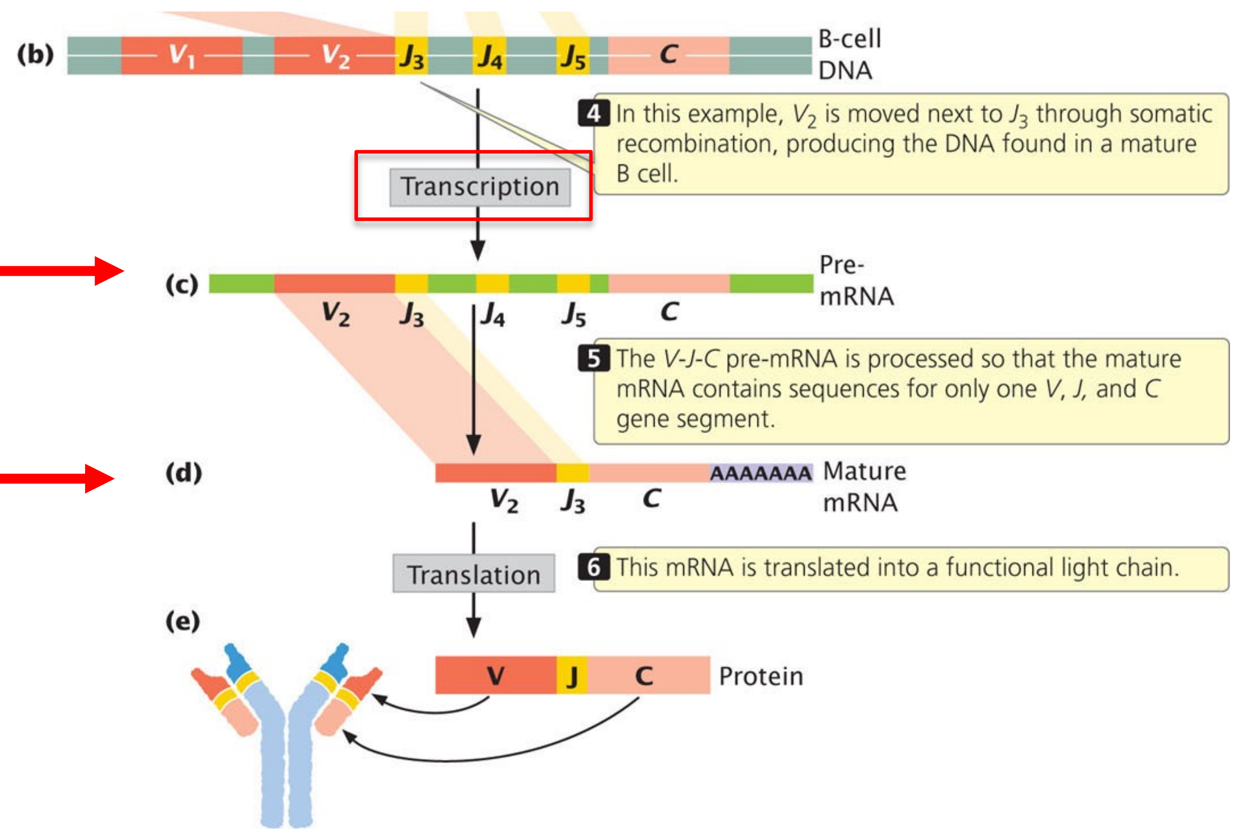
RAG1, RAG2 and DNA repair enzymes introduce dsDNA breaks and joint random V and J segments. **This occurs at the level of DNA.**

Pre-mRNA is spliced to produce unique combination of V-J-C light chain

Primary transcript (pre-mRNA)

Mature RNA (mRNA)



Note that the immunoglobulin generated in a given B cell will always be the same

Pierce. 22.23

The heavy chain locus has multiple D (diversity) gene segments in addition to V J and C segments.

Antibody diversity produced by VDJ segments

	Kappa Light Chains	Lambda Chains	Heavy Chains
Variable (V) segments	30	35	100
Diversity (D) segment	0	0	23
Joining (J) segment	5	7	5
Constant region	1	1	5*
Antibody specificities	150	245	11,500

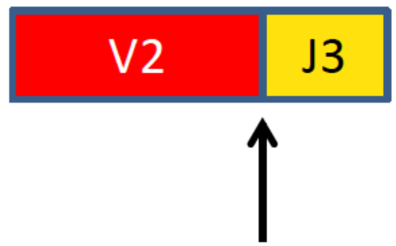
Kappa+ Heavy 1.7×10^6

Lambda+ Heavy 2.8×10^6

Total of 4.5×10^6

* While V, D and J segments contribute to the antibody diversity, the constant regions do not.

Junctional diversity



Imprecise junction

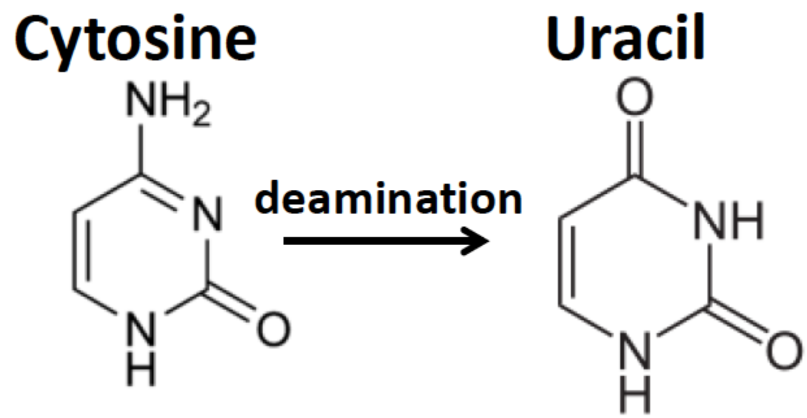
Few random nucleotides are lost or gain

In many cases, it will result in a frameshift that produces a nonfunctional gene.

Somatic hypermutation

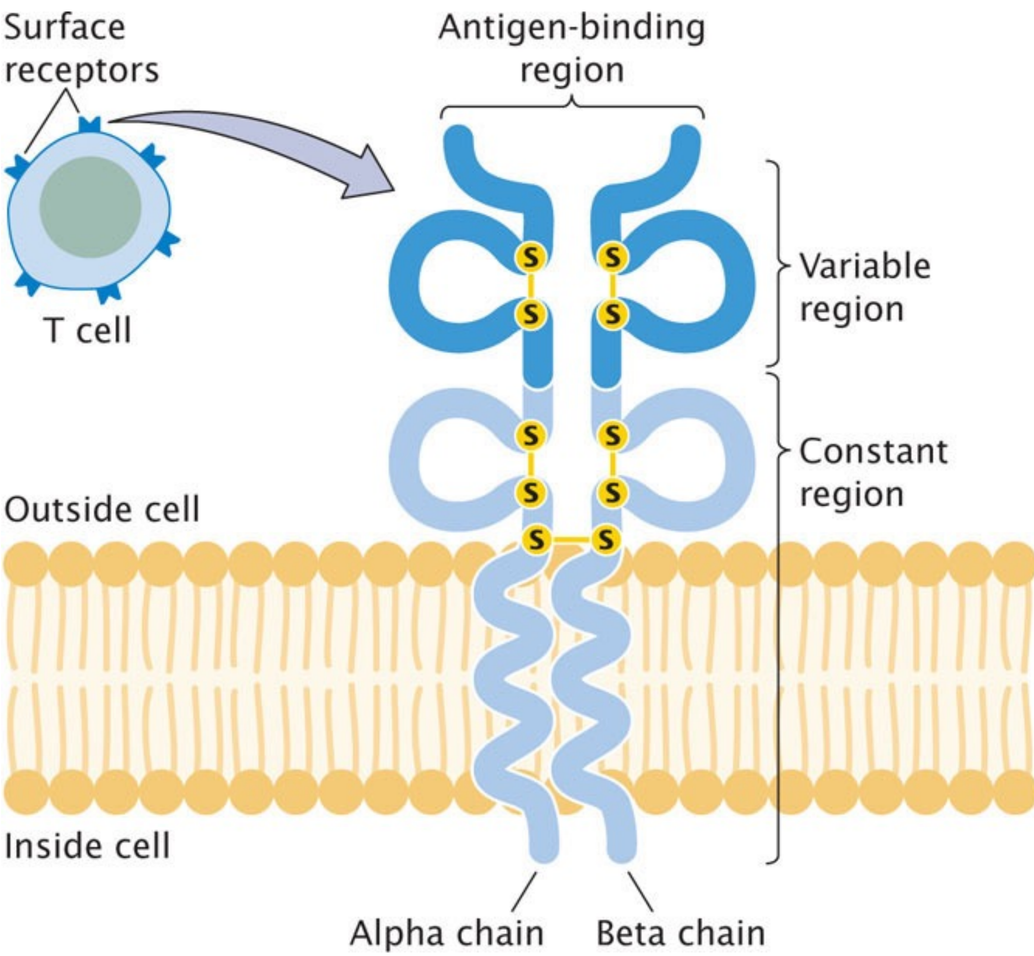
The immunoglobulin genes are subject to a high mutation rate

(deamination of cytosine which become uracil)



Uracil is replaced by the DNA repair mechanisms by another base, thus generating a point mutation

T-cell receptor diversity



Pierce. 22.24

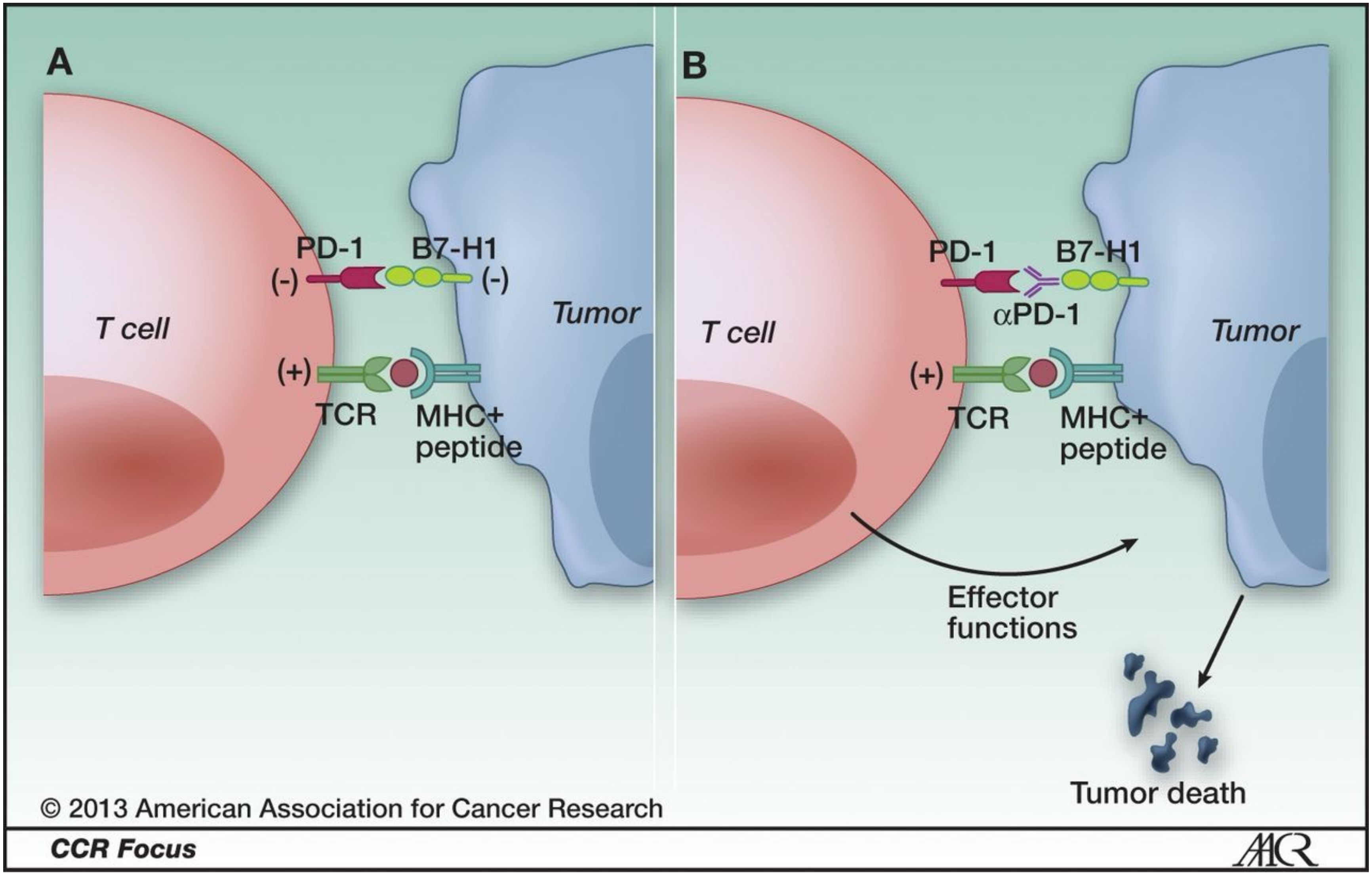
T cell receptors are composed of alpha and beta chains that have variable regions.

Alpha chain: 44 – 46 V gene segments, 50 J gene segments and a single C segment.

Beta chain is similar to Alpha chain, but contains D gene segments.

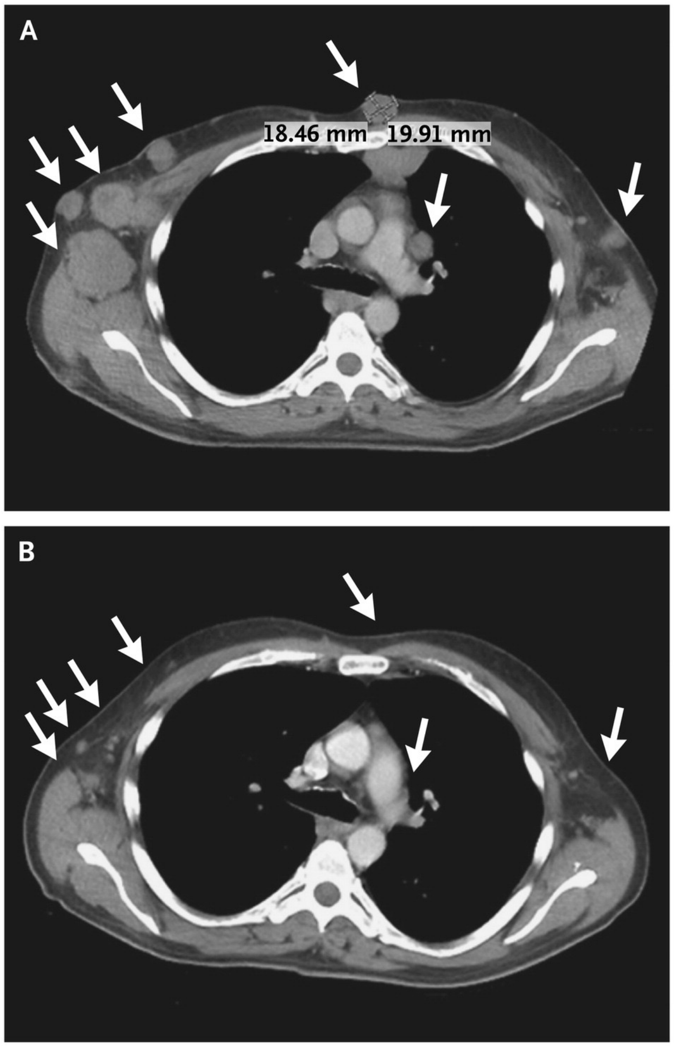
Somatic recombination and junction diversity, **but no hypermutation.**

PD-1 activation by tumor cells suppresses T cell function



Clinical Cancer Research
AACR American Association for Cancer Research

Chest showing tumor regression in a patient treated with CTLA-4 and PD-1 antibodies



Before

4 weeks after treatment

Wolchok JD et al. N Engl J Med 2013;369:122-133.